THE EVALUATION OF DIFFERENT ATTACHMENTS USED FOR MANDİBULAR TWO İMPLANT-SUPPORTED OVERDENTURE PROSTHESİS

MANDULADA İKİ İMPLANT DESTEKLİ OVERDENTURE PROTEZLERDE KULLANILAN FARKLI BAĞLANTILARIN DEĞERLENDİRİLMESİ

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ÖZET

Dişsizlik hastayı fonksiyon, estetik ve psikolojik olarak olumsuz etkilemektedir. Dişsiz maksilla ve mandibulayı rehebilite edip fonksiyon kazandırmak için kullanılan geleneksel yöntem tam protezlerdir. Mandibulanın maksiller çeneden farkı dil ve kas yapılarıyla komşulukların yanı sıra alveolar sırt rezopsiyonunun devam etmesi tam protezlerin stabilitesi ve tutuculuğunu azaltmaktadır. Hastalar genel olarak konuşurken ve yemek yerken alt protezin hareket etmesinden şikayetçidirler. Bu tür hastalar için en yaygın ve ekonomik tedavi seçeneği mandibulaya iki implant destekli overdenture protezlerin yapılmasıdır. İmplant destekli overdenture protezler geleneksel tam protezlere kıyasla artmış retansiyon, stabilite, hasta menmuniyeti ideal ve etkin çiğneme sunmaktadır. Bu protezlerde bağlantı ara parçalarla sağlanmaktadır. Overdenture protezlerde bağlantı sistemleri splintlenen ve splintlenmeyen bağlantılar olarak sınıflandırılabilinir. Splitlenen tutucular bar yardımıyla birbirlerine bağlanarak aşırı rezorbe kretlerde interark mesafenin fazla olduğu durumlarda kullanılmaktadır. Spitlenmeyen tutucular ise birbirinden bağımsız implant üzeri ataşman sistemleridir. Farklı ataçman sistemlerinin avantaj ve dezavantajları mevcuttur. Klinik başarı için uygun ataşman seçimini etkileyen faktörleri ve vakaya özgü kullanım alanlarını bilmek protezin başarısını etkilemektedir. Bu derlemenin amacı kullanılan iki implant destekli overdenture protezlerde kullanılan farklı ataşman sistemlerinin doğru seçimini dolayısı ile protetik başarı oranını, protez bakımını, hasta memnuniyetini, biyolojik dokularla ilişkisini, biyomekanik bağlantılarını ve maliyetini değerlendirmektir.

Anahtar Kelimeler: Overdenture, implant ataşmanları, mandibula, dental implant

Abstract

Edentulism negatively affects the patient functionally, aesthetically, and psychologically. Complete dentures are used to rehabilitate and restore the edentulous maxilla and mandible. The

difference between the mandible from the maxillary jaw, its neighborhood to the tongue, muscle structures, and the continuation of alveolar ridge resorption reduces the stability and retention of complete dentures. The difference between the mandibular jaw from the maxillary jaw, a neighborhood with tongue and muscle structures, as well as continued alveolar ridge resorption, prevents the stability and retention of complete dentures. Patients generally complain that the lower prosthesis moves while talking and eating. The most common and economical treatment option for such patients is to make two implant-supported overdenture prostheses in the mandible. Implant-supported overdenture dentures provide increased retention, stability, patient satisfaction, and ideal and effective chewing compared to conventional complete dentures. Attachment systems in overdenture prostheses are spacers that provide implant and prostheses connection. Attachment systems in overdenture prostheses can be classified as splinted and non-splinting attachments. Splinting attachments relate to a bar and are used in overly resorted crests and cases where the inter arch distance is large. Non-splinting attachments are independent attachment systems on implants. Different attachment systems have advantages and disadvantages. It is useful to know the factors affecting the selection of the appropriate attachment for clinical success and the case-specific usage areas that affect the success of the prostheses. This review aims to evaluate the correct selection of different attachment systems used in two implant-supported overdenture prostheses in the mandibular, thus prosthetic success rate, prosthetic care, patient satisfaction, relationship with biological tissue, biomechanical connections, and cost.

Keywords: Overdenture, implant attachments, mandible, dental implant.

1.Introduction

The number of individuals seeking dental care is rising along with the average lifespan of humans. Conventional complete dentures are typically used to treat edentulous patients' mastication, speech, and aesthetic issues, but unfortunately, due to inadequate retention, support, and stability of their dentures, these patients may not be content with complete dentures. Most patients experience stability issues with their prostheses due to resorbed residual ridges and weakened neuromuscular control. Mandibular overdentures have reportedly produced outcomes comparable to implant-supported permanent prostheses over the long run. For edentulous individuals, the preferred approach is a complete maxillary denture in opposition to a two-implant-retained mandibular overdenture [1]. Compared to conventional complete dentures, this approach greatly improves patient satisfaction and quality of life [2].

Mandibular implant overdentures have improved retention and stability by using several types of attachment methods. These include the bar, magnet, stud (ball or low-profile), LOCATOR attachments, telescopic attachments, and Novaloc systems [1,2]. They each have their advantages and disadvantages.

A bar attachment enables splinting and wide load distribution for the immediate loading procedure, reducing the implants' moment. This also helps correct misaligned implants. However, it has been

noted to be more costly and technique sensitive. In addition, bars provide a problem for individuals with constrained restorative space because they demand more vertical and buccolingual areas. Bars can invade the space for the tongue in patients with tapered arches therefore, they should not be used on individuals with a V-shaped ridge. Gingival hyperplasia is possible underneath the bar since it is challenging to eliminate plaque and practice good hygiene [2,3].

The ball attachment is simple to use and maintain, economical has an extensive range of motion, ideal retention, and has high patient satisfaction [3]. As a result of their low profile, stud attachments can be employed in situations where there is a lack of inter arch space and to maintain good oral hygiene [2]. It comprises an assembly (male component) attached to the implant, an intermediate with a spherical socket at its end, and a female component with a retaining ring connected to the prosthesis' base [4].

Another type of attachment system uses magnets. For more than 60 years, magnetic attachments have been utilized to keep dentures in place [3]. They are inexpensive, and manufacturing and repairing prostheses is simple. In more recent designs, enclosed rare-earth magnets minimize attachment size, boost retentive forces, and lower wear and corrosion susceptibility. Magnets have drawbacks, such as limited resistance to lateral pressures, which leads to an immediate loss of retention. Nevertheless, this has been viewed as a benefit in sheltering the implant from negative lateral stresses [2]. Other complications include corrosion and loss of magnetism [3].

Dentures have been attached to prepared teeth for many years using the double-crown (telescopic)attachment. It offers good retention and stability and is simple to clean. The telescopic attachment's self-seating function makes denture insertion easier and makes it suited for people with limited manual dexterity, such as those with Parkinson’s disease. The double-crown attachment does have certain drawbacks, though, including the appearance of the metal attachments from under the prostheses and the requirement for adequate interarch space due to its high vertical profile [3].

A worldwide survey found that the LOCATOR® system is the most widely utilized stud-type attachment mechanism for implant-supported overdentures [5]. Due to the LOCATOR attachment's low 2.5-mm height, self-aligning design, which can adjust for implant angulations of up to 40 degrees, and absence of denture base deformation or fracture even when utilized in patients with limited inter arch space, it has gained popularity over the past 20 years [3]. However, there have been several reports of excessive wear, loss of retention, higher maintenance needs, a deterioration in retentive capacity in instances with nonparallel implant divergences, and various contradicting clinical behaviors [5].

A novel form of attachment called Novaloc is based on mechanical retention from a polyetheretherketone (PEEK) matrix on a cylindrical patrix and may be more durable than the nylon used in previous systems, it is an alternative to these attachment methods. [6]. Individual attachments, such as the Locator system (ZestAncors, Escondido, CA, USA), are clinically utilized for significantly divergent or converging implants. However, in these circumstances, the retention

performance can only be somewhat altered by using particular matrices (Locator Replacement Males), and a significant angulation compensation at the attachment level cannot be guaranteed. The Locator system exhibits higher wear behavior in the form of repair and maintenance susceptibility, despite producing favorable retention behavior outcomes in vitro studies. Systems with angled abutments (patrices) are becoming more widely available on the market as a solution to this problem. For instance, in addition to straight abutments, the Novaloc Retentive System for hybrid dentures (Institut Straumann AG, Basel, Switzerland) includes an abutment inclined at 15 degrees. Additionally, these abutments' abrasion-resistant surfaces, which are made of amorphous diamond-like carbon (ADLC), are designed to improve the wear behavior [7].

This study aims to evaluate the different attachment systems used for two implant-supported overdenture prostheses and compare them under different clinical categories.

2. Materials and Methods

2.1 Search strategy and information sources

Electronic searches were carried out on PUBMED/MEDLİNE, Scopus, Cochrane, and Google Scholar databases for eligible articles published up to October 2022 concerning the aim of this study. The search was limited to articles published in the last five years. The keywords used for the search were: ‘‘mandible, overdenture and two implants and attachments’’. The keywords were added to limit the search. When just the keyword ‘mandible’ was used, 12,817 results were found. With the addition of ‘overdenture,’ this number dropped to 371. Adding on the keyword ‘two implants’ dropped the results to 214. Finally, when the keyword ‘attachments’ was added, a total of 116 results remained.

2.2 Eligibility criteria

For these final results, the inclusion criteria were as follows: studies with at least 10 participants and a minimum follow-up period of 6 months, studies published in English, and studies that allowed the collection of the necessary information. The exclusion criteria were studies with mini-implants, studies on the maxillary jaw, case reports, and animal studies. This systematic review analyzed 36 articles that included information relevant to this study and met these criteria.

The PICO (Population, Intervention, Comparison, and Outcome) strategy was used to answer the question: ‘‘Are there significant clinical, biological, and mechanical differences between different attachment systems used for two implant-supported mandibular overdentures?’’. Population: Patients that have two implant-supported overdenture prostheses,

Intervention: Patients with different attachment systems,

Comparison: Comparing the different attachment systems used for overdenture prostheses,

Outcome: Evaluation of clinical, biological, and mechanical success between these different attachments.

2.3 Article Review Process

The reviewers evaluated the articles that were searched. All identified papers' titles were reviewed to choose the number of Abstracts to review. In the following phase, article abstracts were scrutinized to see if the study's contents would be relevant to this review's goal. The reviewers agreed upon the qualifying list of papers for the following round of review. After reading the entirety of every item that met the qualifying requirements, those that did not meet the criteria were disqualified.

3. Results

At the end of the selection process, a total of 36 articles were analyzed and included in this analysis. Seven of these articles concluded that there were no significant differences between the different attachment systems that were analyzed. Değirmenci et al. split 53 patients into six groups, in which a mandibular overdenture supported by a bar attachment was favored by 54.72% of the participants. In contrast, a ball attachment was utilized by 45.28% of the participants. Ball or bar attachment choice did not significantly impact the patient's seven categorical assessments, according to the Mann-Whitney U test ($p>0.05$). Patients with bar attachment had lower ratings for physical pain, psychological discomfort, physical impairment, psychological disability, and social disability. But this distinction lacked statistical significance [8]. Similarly, Varshney et al. also compared the ball and bar attachments. They concluded that there was no substantial difference between the attachment systems regarding implant failure, replacement of attachment fragments, and broken overdentures [9]. In a meta-analysis, 380 patients with an average age of 62.8 years received a total of 984 implants. In terms of minor bone loss (P =.39; MD: 0.11; 95% CI: 0.37 to 0.14); complications (P =.31; RR: 1.26; CI: 0.80-1.99); and implant survival rate (P =.14; RR: 0.37%; CI: 0.10-1.36), the meta-analysis found no statistically significant differences between splinted(bar) and unpainted (balls/O-rings, Locator, ERA, magnets, and crowns) attachment techniques [10]. Reem M Abdeen et al. studied the effects of attachment systems on the bone density around the implants of a mandibular overdenture and found that the difference was negligible [11]. A systematic review found that conventional ball, mini ball, and low-profile attachments have little effect on PROMs (Patient-Reported Outcome Measures) when there is a typical inter arch space. Low-profile attachments exhibited better PROMs in a narrower inter arch space [12]. Regarding peri-implant health, there appeared to be no evident superiority of one attachment type over another [13]. Rosa et al. compared the patient satisfaction of users of the ball attachment versus the other attachment systems and concluded that there was no significant difference [4].

In contrast to these results, nine articles evaluated in this systematic review stated a significant difference between different attachment systems. Sutariya et al.'s outcomes were that bar attachment provided the most superior retention, the ball attachment system is an advantageous option for limited inter arch space and parallel implant placement, probing depth and marginal bone loss did not statistically significantly differ between the bar and telescopic attachment, and ball attachment reported fewer complications than locator attachment [14]. A systematic review by Bansal et al. showed that overdentures with ball attachments demonstrate substantial retention, survival, and overall performance. In contrast, those involving bar attachments result in mild tissue alterations and bone resorption. Under functional load, the magnetic attachments reported more bone resorption. They also concluded that the locator attachments need regular maintenance and repairs [15]. A 5-year randomized controlled clinical trial showed that For 2-implant overdentures in patients with atrophied mandibles, bar attachments showed the highest incidence of complications related to mandibular overdentures(overdenture relines, fractures, hyperplasia, and soreness), stud attachments showed the highest incidence of complications related to the attachment components(wear and replacement of retentive components), The telescopic group showed the highest incidence of screw loosening, teeth wear/fracture, fractures, teeth wear, teeth separation, and flappy ridges [16]. The effects of loading protocols and different attachment systems on peri-implant health and prosthetic complications were analyzed by Aldhohrah et al. Similar effects on peri-implant health were seen with all overdenture attachment methods, whether they were immediate or delayed loading. Ball+ delayed loading revealed the least probing depth, whereas bar+ instant loading was associated with the least minimal bone loss [3]. Prasad et al. concluded that prosthetic complications were least common for locator attachments and most common for magnet attachments [2]. In one study, the biomechanical properties of three distinct types of magnetic attachments in two-implant-retained mandibular overdentures were compared using a three-dimensional finite element analysis. Magnetic attachments of the flat, dome and cushion types were created to hold the overdenture in place. The results of two-implant-retained mandibular overdentures revealed that cushion-type and dome-type magnetic attachments are preferable [17]. A Randomized Controlled Clinical Trial compared bar (BOD), telescopic (TOD), and Locator (LOD) Attachments. The greatest plaque scores, gingival scores, and pocket depths were recorded by BOD, followed by LOD, while TOD reported the lowest values. There was no discernible difference between groups in terms of implant stability or keratinized mucosa. The greatest horizontal and vertical bone losses were seen in TOD and BOD, respectively. Vertical and horizontal bone losses were the lowest for LOD. Between groups, there was no difference in the implant survival rate [18]. In H. Sato et al.'s study, the effects of attachment types, Locator attachments (LA), ball attachments (BA), magnetic attachments (MA), and round-bar attachments (R-BA), on the pressure exerted on the oral mucosa were investigated. BA appeared to be effective for lowering oral mucosa pressure during mastication since it has the greatest impact on support and bracing [19]. One Crossover Trial analyzed different Attachments Used for Implant-Retained Overdentures and the Oral Health-Related Quality of Life (OHRQoL) in Subjects with Resorbed Mandibles. Bar, telescopic, and stud attachments were used for this study. In terms of self-

consciousness and the perception that the prosthesis is an integral part of the patient, studs are preferable. While telescopic and stud attachments were more successful with oral health, bar and telescopic attachments were more reliable while biting into soft and hard meals [20]. A retrospective study showed that the ball attachment had a lower gingival index and pocket depth (p 0.05). Ball attachments were more frequently associated with prosthetic issues such as denture fracture and attachment wear. Patients who had bar attachments required substantially fewer post-insertion and attachment replacement appointments (p 0.05) [21]. However, Chrcanovic et al. concluded that the bar-clip system had a higher prevalence and rate of complications than the ball attachment method, indicating that overdenture's using the bar-clip system could need more chair time and visits for adjustments, which would raise the patient's maintenance expenditures. Single attachments like the ball system are typically claimed to be simpler to maintain or repair [22]. An in vivo study's findings matched these results and stated that compared to the ball-and-socket attachment, the bar-and-clip attachment had higher Modified Sulcular Bleeding Index and Plaque Index values [23]. Two studies comparing the ball and bar attachments stated that there was no statistically significant difference in patient satisfaction ratings between the ball and bar attachments. Neither the ball attachment nor the bar attachment was preferred over the other [24,1].

Six of the studies examined compared the locator(stud) and ball attachment systems. D'haese et al. compared the type, amount, and total cost of required maintenance and concluded that regardless of the kind of retention, the maintenance cost up to seven years is correspondingly less than 20% of the original treatment cost. However, because of several outliers, there is a wide range. Due to wear, the locator requires more visits, but the overall cost is likely to be similar because the cost of these small plastic matrices that usually need replacing is not very high [25]. Different materials, such as PEKK and nylon, can be used for these stud attachments. One article compared the retentive features of these different materials and concluded that the PEKK attachment system showed less abrasion and change in retention than the nylon [26]. A systematic review and meta-analysis showed as choices for mandibular overdenture retainers, stud, and ball attachments are both clinically acceptable. However, the stud demonstrated improved satisfaction, reduced prosthesis aftercare, and marginal bone loss [27]. Abdelaziz et al.'s study inspected the difference in retention between these two attachment systems and found that the Locator R-TX lost 26.31% of its retention over a simulated 5-year period of activity, while the ball and socket lost 19.87% of its retention. Locator F-TX lost even more overall retention than Locator R-TX [28]. Wakam et al. studied the retention, wear, and maintenance effects of cylindrical attachment systems (CAS) and ball attachment systems (BAS). The study showed that CAS should be chosen in the following situations: (i) In situations of low vertical prosthetic space, although they have a greater cross-section than the 2.25 mm standard diameter for BAS, (ii) when initial retention is required to be higher (for example, in lingual parafunction, bruxism, or in cases of a poor surface of sustentation) and (iii) if the implant's axis is different from the insertion axis of the prosthesis by more than 30°. BAS should be selected: (i) When there is a low bone width and (ii) the necessary retention is not strong but steady over time (e.g., if the patient has difficulty making frequent visits to the practitioner or is disabled) [29]. The effect of attachment systems on the strain of complete

mandibular overdentures was investigated in a study that revealed that under unilateral and bilateral loading, there is a considerable increase in strain for the ball compared to the locator but not significantly so in the posterior region [30]. Finally, cost-effectiveness was analyzed by Matthys et al., and this study showed that Locator attachment systems were more cost-effective [31].

A 2- to 12-year retrospective analysis inspected the following characteristics of bar attachments were evaluated: fractures of acrylic resin (base fractures, midline fractures), debonding of teeth, fractures of opposing prostheses, the requirement for relining or rebasing, fractures of abutment and bar screws, wear on ball or bar attachments or clips, fractures or detachments, fractures of bars, and implant loss. The Rhein 83 polymer attachment wearing out was the most frequent issue, followed by bar screw loosening. At 6.8 years, the total overdenture survival rate was 91.9% [32]. A different study compared the retention and loss of retention between two types of bar clip materials (digitally created PEEK bar clip and standard Nylon bar clip) following fatigue testing at various time intervals. At the start of the trial, there were statistically significant differences in retention between the PEEK and nylon bar clips (p=0.000*). However, there was no discernible difference in retention between the test groups after three years of simulated usage (p=0.055, NS) [33]. A 5-year retrospective study comparing bar and magnetic attachment systems concluded that the magnetic attachment group considerably outperformed the bar attachment group in terms of peri-implant probing depth (PPD) and plaque index (PI) (P0.05). Complications with prosthetics and upkeep depended on the attachment. Most of the participants that were recruited were happy with their prostheses. Between the two groups, there was no statistically significant difference in OHRQoL (Oral Health-Related Quality of Life) or overall patient satisfaction (P>0.05). However, patients commented that the Bar attachment overdentures were much more challenging to clean than the magnetic attachment overdenture [34]. Dhamodaran et al. performed an in vitro study comparing the retention of bar and locator attachments. The results demonstrated that compared to the locator attachment method, the bar and clip attachment system provided superior retention [35].

Arnold et al. compared the retentive characteristics of new systems. Novaloc was compared to the locator attachment system. All the evaluated attachment devices exhibited wear responses during the simulated wearing time of more than nine years. The Novaloc attachment method displayed a comparably continuous curve with a retention force loss of 26.14%. However, the Locator attachment system's overall retention force loss, which was 77.5%, was within the clinically significant range [7]. Shehata et al. compared the Novaloc system to telescopic attachment systems. At baseline and following exposure to various cyclic loads, retention of group I (Telescope) was substantially greater (p 0.05) than those of group II (Nova loc) (500- 1000- 1500- 2000). In terms of wear, a t-test between mean differences was used to compare wear (wear change) between the two groups, and the results showed that group I (the telescope) had considerably more wear than group II (Nova-loc). The retentive forces of the Telescope and Nova-loc systems are adequate; nevertheless, implant overdentures have had short-term clinical

success. In conclusion, the retention value of Telescopic attachment systems is more significant than that of Novaloc. However, Novaloc exhibits a lower level of wear because of its peek material [36].

4. Discussion

According to the McGill Consensus in 2002 (Feine et al., 2002) and the York Consensus Statement with McGill, overdentures supported by two implants are thought to be the best option for treating entirely edentulous individuals (Thomason et al., 2009) [4]. The retention of the remaining alveolar ridge, improved retention and stability, chewing efficiency, enhanced comfort and happiness, and improved quality of life are only a few benefits of implant overdentures [35]. Due to the advantages that it can provide, the use of mandibular implant-supported overdentures has become more common; therefore, studies in this field have gained importance. One of the significant decisions made for planning and designing an overdenture is the choice of the attachment system that should be used.

The choice of attachment type is influenced by several variables, including bone height, bone breadth, inter arch space, level of retention necessary, patient economic situation, patient anticipation for prosthetics, and doctor preference. Between the mandibular incisal edge and mucosa, there should be a minimum of 13–14 mm available for rehabilitation for bar attachment, 13–14 mm for telescopic attachment, 10–12 mm for ball attachment, 8.5 mm for locator attachment, and 8.5 mm for magnet attachment. Ball attachment can withstand an angle of 10 degrees or less between the implants. While locating attachments can readily correct angulations up to 40 degrees. When utilizing the telescopic attachment, the parallelism of the implants is of utmost importance. With the use of angulated abutments, bar attachment may handle no parallelism. However, implant-supported overdenture is linked to problems in the peri-implant soft and hard tissues, frequent follow-up visits, prosthetic failure, and overdenture attachment loosening. The manifestation of these issues may change depending on the attachment [14]. Hence the right choice of an attachment system becomes essential.

All the studies that compared the retention of attachment systems were compiled, and the results are as follows. Varshney et al.'s study is the only one that suggests there is no significant difference between the different attachments [9]. Abdelaziz et al.'s results indicate that the ball attachment is superior to the locator attachment, while Wakam et al. further expanded on this topic, saying that the locator provided better initial retention, while the ball attachment provided more stable retention [28,29]. However, Gonçalves et al. stated that the bar attachment's initial retention is better than the ball [24]. This could be because the bar and clip provide a higher surface area for retention. An in vitro experiment revealed that the locator attachment system was weaker than the bar and clip attachment in terms of retention [35]. A new attachment system that is coming into use is the Novaloc system. An in vitro study concluded that telescopic systems had better retentive qualities than the Novaloc, and Arnold et al. state that Novaloc was significantly more retentive

than the locator [36,7]. This is thought to be due to the new and improved material usage in the Novaloc system (PEKK).

Regardless of the kind of attachments, implant overdentures for those with mandibular bone resorption increased patient satisfaction and OHRQoL compared to conventional dentures. 13 of the articles analyzed for this review discussed the patient's satisfaction in one way or another. 5 of these stated that there was no significant difference in the comparison of the patient's satisfaction [1,4,12,24,34]. A crossover trial's data suggests that studs are more favorable in terms of comfort with self-consciousness and the perception that the prosthesis is a part of the patient. While telescopic and stud attachments provided more oral hygiene satisfaction, bar and telescopic attachments provided greater gratification during chewing soft and hard meals [20]. Two studies concluded that ball attachments had the most complications [14,21]. However, Chrcanovic et al.'s data suggest that the bar-clip system had a higher prevalence and rate of complications than the ball attachment system, indicating that overdentures using the bar-clip system could need more chair time and visits for adjustments, which would raise the patient's maintenance expenditures. Additionally, splinted attachments like the bar-clip system are typically believed to be more difficult to maintain or repair than single attachments like the ball system [22]. D'haese et al. stated that due to the wear on the locator's nylon features, this system requires more frequent visits to the dentist [25]. Nevertheless, other studies proved that the locator system has the least number of complications and is the most cost-effective [2,27,31].

Another parameter to take into consideration is the level of oral hygiene. This includes variables such as marginal bone loss, peri-implant health, bleeding index, and plaque index. Four articles concluded that the attachment system did not affect the above parameters [10,13,14,24]. When compared to alternative attachments, telescopic attachments were related to better clinical peri-implant soft tissues. Locator attachments, however, could help maintain the peri-implant bone. Bar attachment systems recorded the highest plaque scores, gingival scores, and pocket depths [18]. Other studies showed similar results stating that the bar-and-clip attachment exhibited higher Modified Sulcular Bleeding Index and Plaque Index values [23,34]. This is most probably because the bar and clip system is difficult for the patient to clean. Food remnants and plaque can easily invade the space between the gums and the bar.

Some limitations of this study include that the analyzed data came from studies with only a 1-year follow-up; these findings should be viewed with care. Therefore, more RCTs comparing the various overdenture attachments are still required. These RCTs should have bigger sample sizes and more extended follow-up periods. A limited number of relevant studies about the Novaloc system could be included as this is new, and the research around it is not as encompassing as the other attachment systems. It is also a complicated process trying to compare parameters such as complications because each overdenture attachment has a unique set of complications; therefore, the present study's findings on prosthetic issues were ambiguous.

5. Conclusions

1. The Novaloc system provides better retention than the locator system due to the use of polyetheretherketone (PEEK), making it more durable and less prone to wear.
2. There is no significant difference between patient satisfaction with overdentures with different attachment systems.
3. Bar attachments make it challenging to eliminate plaque and practice good hygiene, therefore, they can cause adverse periodontal effects.

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